

K9SAR Technical Reference

June 2026

SUMMARY

K9SAR is a comprehensive Search and Rescue K9 certification, evaluation, administration, and public information management system developed for Tri-State K9 and affiliated organizations.

TECHNICAL SKILLS

React
Vite
TypeScript
TailwindCSS
FastAPI
SQLAlchemy
JWT Authentication
MySQL
AWS Lightsail
Linux VM
REST APIs
PDF generation
ID card rendering
Public verification services
Forum and survey system

PURPOSE

The system manages handlers, dogs, teams, certifications, standards, evaluators, public verification, identification cards, forums, surveys, and organizational administration.

This document serves as the primary technical reference for future administrators, developers, and maintainers.

SYSTEM ARCHITECTURE

Frontend communicates exclusively through REST APIs provided by the backend.

Backend responsibilities include business logic, authorization, certification workflow enforcement, signature verification, PDF generation, ID card rendering, and public verification services.

Database is the system of record with major entities including Users, Handlers, Dogs, Teams, Certifications, CertificationEvents, Standards, Disciplines, DisciplineGroups, Affiliations, Forums, and Ballots.

CORE DATA MODEL

User: authenticated system account with roles assigned through user_roles. Handler: SAR handler linked to affiliations, teams, and certifications through teams. Dog: search dog; may exist independently of teams; orphan dogs removable through administrative cleanup. Team: handler-dog pairing; certifications always attached to teams for historical tracking.

CERTIFICATION SYSTEM

Certifications are immutable historical records; updates generally create CertificationEvents.

Statuses: ACTIVE, PENDING, INCOMPLETE, REJECTED, REVOKED, EXPIRED.

Multipart modes: NEVER, ALWAYS, FIRST_CERT_ONLY.

Co-evaluator workflow supports PENDING to ACTIVE/REJECTED transitions with limited issuing evaluator edits while pending.

STANDARDS SYSTEM

Hierarchy: DisciplineGroup → Discipline → Standard. Display ordering via DisciplineGroup.sortorder and Discipline.sortorder. Standards stored as uploaded PDF documents at /api/documents/file/{hash}.pdf. Booklet endpoints include /api/standards/print/standards-booklet and /api/standards/print/standards-summary-booklet.

SIGNATURE AND INTEGRITY SYSTEM

User signatures stored in user_signatures with image_png and sha256_hash. Certificates include cryptographic seal generated using build_seal_payload_v2(). QR verification recalculates seal with results VALID, TAMPERED, UNKNOWN.

ID CARD SYSTEM

Card types: Handler ID, K9 Team ID. Layout: 3.5" × 2", 2 × 5 cards, 10 cards per sheet, optimized for Avery 8879 stock. Headshots: HandlerIdHeadshot and DogIdHeadshot. Affiliations may upload background image and branding information; handler cards can display affiliation branding; dog cards remain standardized.

PUBLIC DIRECTORY SYSTEM

Searchable public certified team directories. Endpoint pattern: /api/public/affiliations/{slug}/directory. Modes: all=0 current certifications only; all=1 includes teams without current certifications.

FORUM AND SURVEY SYSTEM

Forums include topics and posts; content preserved when users are deleted. Surveys implemented through ballot objects: ForumBallot, ForumBallotChoice, ForumBallotVote, ForumBallotFeedback. Administrative reporting includes vote counts, percentages, and freeform responses.

ADMINISTRATIVE CLEANUP SYSTEM

Destructive operations require Preview, Confirmation Text, HMAC Validation, and Expiration Timeout. Cleanup audit log table: AdminCleanupEvent records actor, deleted items, affected IDs, warnings, and confirmation text. Delete preview workflow includes count generation, HMAC creation, confirmation phrase, hard delete validation, audit record write, and transaction commit.

AWS INFRASTRUCTURE

Current environment: AWS Lightsail with Linux VM, backend service, frontend static hosting, and MySQL database. Administrative access should use IAM users rather than root credentials. MFA should be enabled for all administrator accounts.

SOURCE CONTROL

Frontend repository: k9sar_frontend. Backend repository: k9sar_backend. Primary branch: main. Production deployments should originate from committed source control changes.

DESIGN DECISIONS AND RATIONALE

Certifications attach to teams (not handlers or dogs) to preserve historical pairings. Forum posts survive user deletion to preserve organizational history. Shared dogs are protected; only orphaned dogs removed during handler deletion. FIRST_CERT_ONLY multipart mode supports standards requiring multipart only for initial certification. Certificates are cryptographically sealed for independent verification.

OPERATIONAL RUNBOOK

Common administrator tasks: Create User, Assign Role, Create Handler, Create Team, Issue Certification, Approve Certification, Generate ID Cards, Generate Standards Booklet, Review Audit Logs, Perform Cleanup. All destructive operations should use preview routes first.

KNOWN FUTURE ENHANCEMENTS

Natural-language reporting. AI-assisted administrative search. Certificate signature snapshot table. Expanded custom GPT knowledge integration. Additional public embed options. Enhanced audit reporting. Full technical API documentation generation.